BMA (BROADCAST MULTI ACCESS)

Router 1 -Initially Router 1 is DR and Router 2 is BDR

later Router2 becomes DR and router 1 becomes DR other after ospf process is cleared in router 1

R1

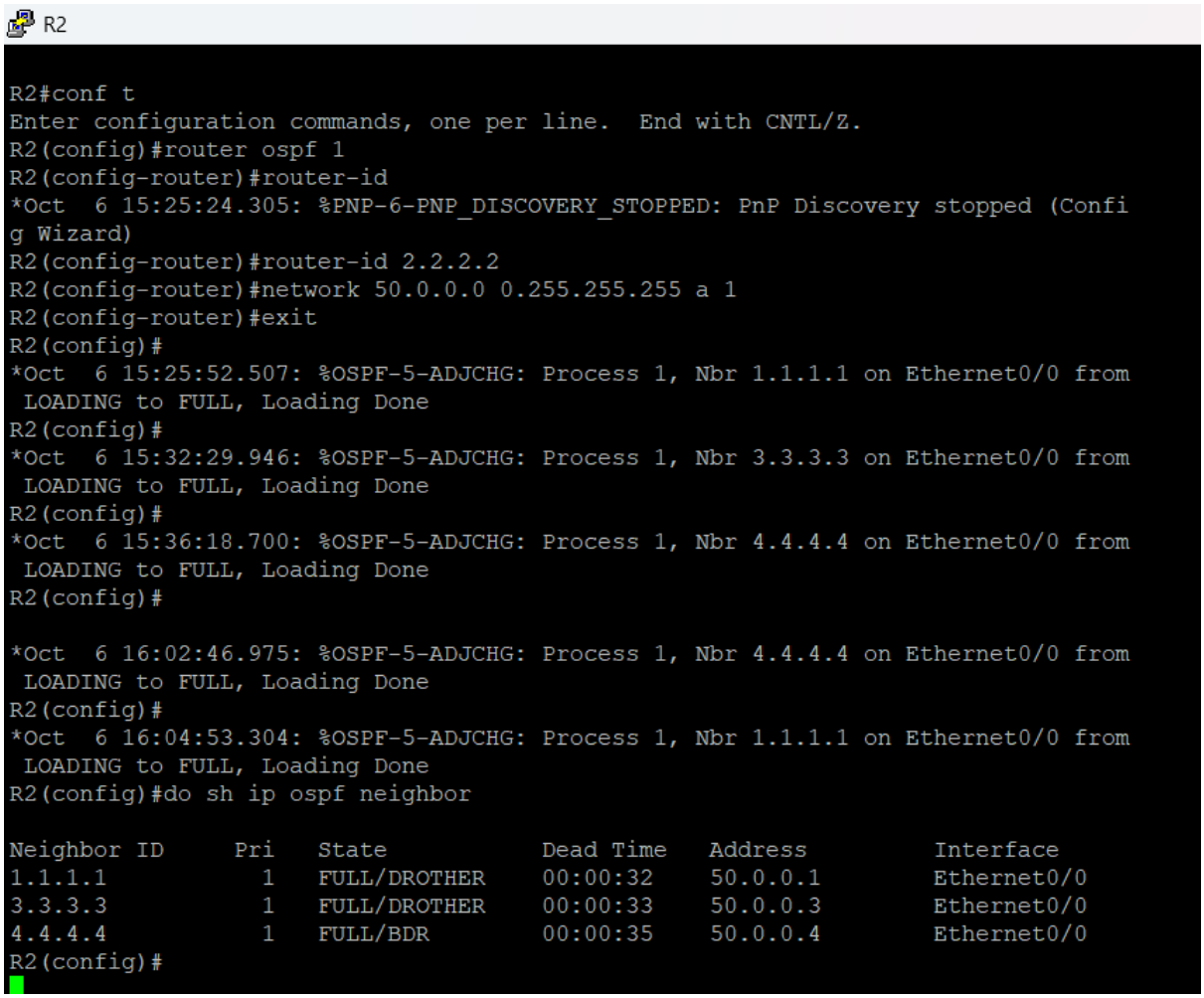
```
R1#rou
% Incomplete command.

R1#
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router ospf 1
R1(config-router)#router-id 1.1.1.1
R1(config-router)#network 50.0.0.0 0.255.255.255 a 1
R1(config-router)#exit
R1(config)#
*Oct  6 15:25:13.693: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Confi
g Wizard)
R1(config)#
*Oct  6 15:25:52.506: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on Ethernet0/0 from
 LOADING to FULL, Loading Done
R1(config)#
*Oct  6 15:32:32.475: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on Ethernet0/0 from
 LOADING to FULL, Loading Done
R1(config)#
*Oct  6 15:36:18.700: %OSPF-5-ADJCHG: Process 1, Nbr 4.4.4.4 on Ethernet0/0 from
 LOADING to FULL, Loading Done
R1(config)#

*Oct  6 16:02:46.975: %OSPF-5-ADJCHG: Process 1, Nbr 4.4.4.4 on Ethernet0/0 from
 LOADING to FULL, Loading Done
R1(config)#do clear ip ospf process
Reset ALL OSPF processes? [no]: yes
R1(config)#
*Oct  6 16:04:38.733: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on Ethernet0/0 from
 FULL to DOWN, Neighbor Down: Interface down or detached
*Oct  6 16:04:38.733: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on Ethernet0/0 from
 FULL to DOWN, Neighbor Down: Interface down or detached
*Oct  6 16:04:38.733: %OSPF-5-ADJCHG: Process 1, Nbr 4.4.4.4 on Ethernet0/0 from
 FULL to DOWN, Neighbor Down: Interface down or detached
R1(config)#router ospf 1
*Oct  6 16:04:52.080: %OSPF-5-ADJCHG: Process 1, Nbr 4.4.4.4 on Ethernet0/0 from
 LOADING to FULL, Loading Done
R1(config)#router ospf 1
*Oct  6 16:04:53.303: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on Ethernet0/0 from
 LOADING to FULL, Loading Done
R1(config)#do sh ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2           1   FULL/DR         00:00:31    50.0.0.2        Ethernet0/0
3.3.3.3           1   2WAY/DROTHER    00:00:31    50.0.0.3        Ethernet0/0
4.4.4.4           1   FULL/BDR        00:00:33    50.0.0.4        Ethernet0/0
R1(config)#
```

Router 2 becomes new DR and Router 4 as BDR



R4

```
R4#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R4(config)#router ospf 1
R4(config-router)#
*Oct  6 15:33:28.585: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Confi
g Wizard)
R4(config-router)#router-id 4.4.4.4
R4(config-router)#network 50.0.0.0 0.255.255.255 a 1
R4(config-router)#exit
R4(config)#do sh ip
*Oct  6 15:36:18.705: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on Ethernet0/0 from
 LOADING to FULL, Loading Done
*Oct  6 15:36:18.705: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on Ethernet0/0 from
 LOADING to FULL, Loading Done
R4(config)#do sh ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           1   FULL/DR         00:00:37    50.0.0.1        Ethernet0/0
2.2.2.2           1   FULL/BDR        00:00:37    50.0.0.2        Ethernet0/0
3.3.3.3           1   2WAY/DROTHER    00:00:32    50.0.0.3        Ethernet0/0
R4(config)#
do ping 50.0.0.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 50.0.0.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R4(config)#do sh ip ospf database

            OSPF Router with ID (4.4.4.4) (Process ID 1)

                Router Link States (Area 1)

Link ID         ADV Router      Age         Seq#       Checksum Link count
1.1.1.1         1.1.1.1         1518        0x80000008 0x00D3E0 1
2.2.2.2         2.2.2.2         1518        0x80000007 0x009715 1
3.3.3.3         3.3.3.3         1523        0x80000003 0x006146 1
4.4.4.4         4.4.4.4         1522        0x80000002 0x00257A 1

                Net Link States (Area 1)

Link ID         ADV Router      Age         Seq#       Checksum
50.0.0.1        1.1.1.1         1523        0x80000003 0x008D43
```

```
R4(config)#
do sh ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           1   FULL/DROTHER    00:00:33    50.0.0.1        Ethernet0/0
2.2.2.2           1   FULL/DR         00:00:33    50.0.0.2        Ethernet0/0
3.3.3.3           1   FULL/DROTHER    00:00:32    50.0.0.3        Ethernet0/0
R4(config)#do sh ip ospf database

            OSPF Router with ID (4.4.4.4) (Process ID 1)

                Router Link States (Area 1)

Link ID         ADV Router      Age         Seq#       Checksum Link count
1.1.1.1         1.1.1.1         1506        0x8000000E 0x00D1DB 1
2.2.2.2         2.2.2.2         1507        0x8000000E 0x009311 1
3.3.3.3         3.3.3.3         1507        0x8000000A 0x005D42 1
4.4.4.4         4.4.4.4         1503        0x80000008 0x002375 1

                Net Link States (Area 1)

Link ID         ADV Router      Age         Seq#       Checksum
50.0.0.2        2.2.2.2         1513        0x80000002 0x005775
R4(config)#
```